

2024 September Floods and Landslides

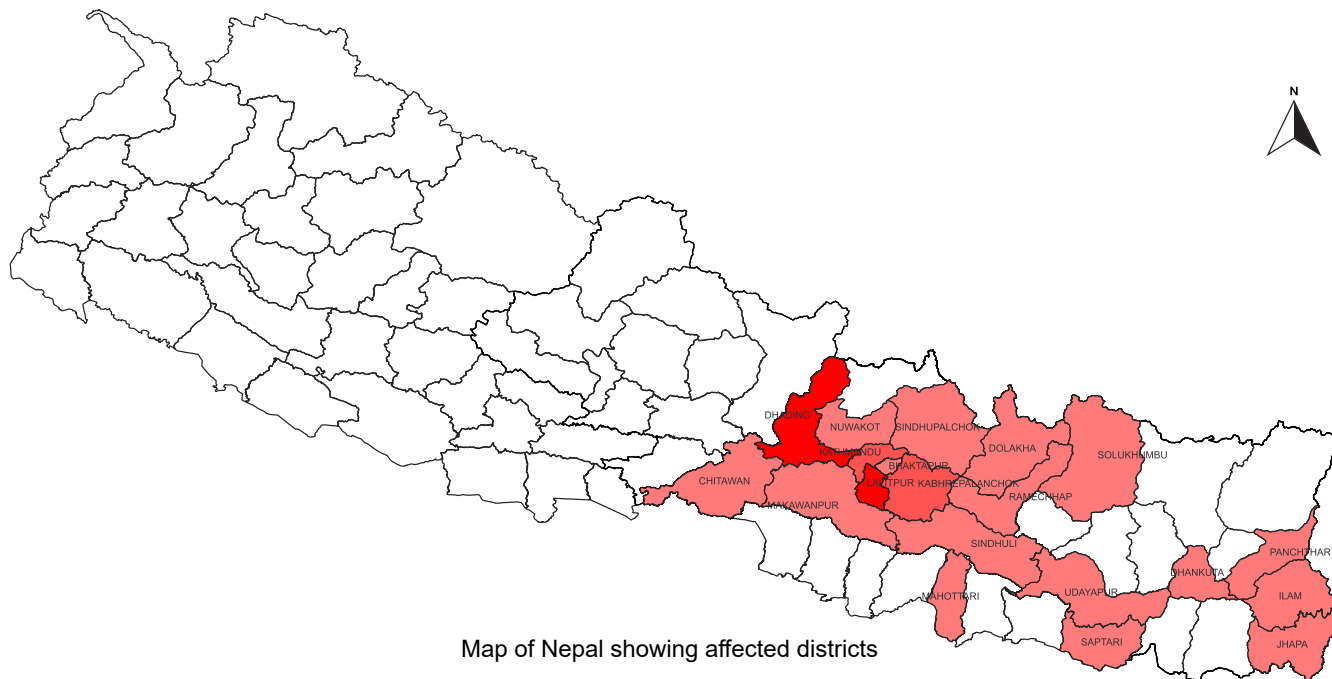
Situation Report #1

(as of 29 Sep, 2024)



Government of Nepal
Ministry of Home Affairs
National Disaster Risk Reduction and
Management Authority

This report is produced by National Disaster Risk Reduction and Management Authority (NDRRMA) in collaboration with sectoral ministries, departments, provincial and local governments. It covers the situation of floods and landslides due to the ongoing heavy rainfall across the country since 26 September.



Map of Nepal showing affected districts

Highlights

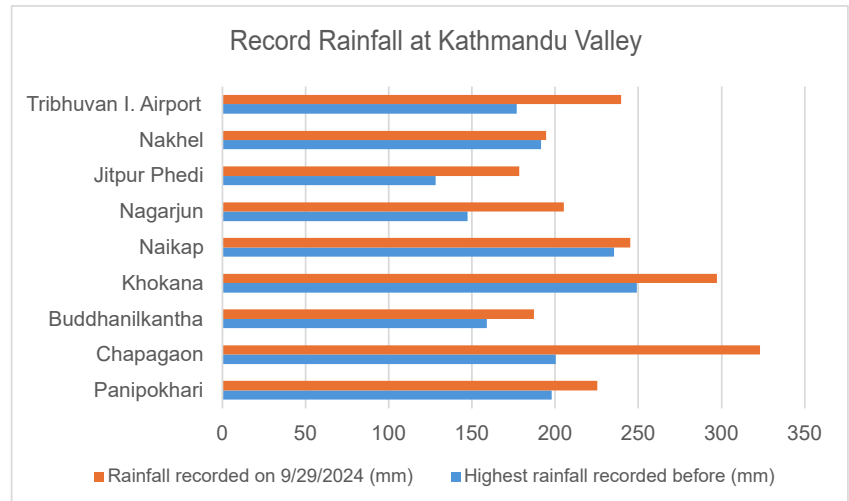
1. A total of 20 districts are reported to be affected by the continuous and record rainfall from 26 to 28 of September.
2. More than 170 people died, 111 people were injured and 42 people are missing in Kathmandu, Bhaktapur, Lalitpur, Koshi, Madesh and Bagmati Province with more 4000 people rescued.
3. 23 highways across the country have been affected, with an estimated loss of over 2 billion due to roadblocks and damaged bridges.
4. The Bhoté Koshi River has swept away three concrete bridges Larcha, Liping, and Ghatte Khola on the Araniko Highway connecting Nepal and China, while landslides have blocked the road at numerous locations, and water levels in various rivers, including the Narayani, Kankai, Kamala, Bagmati, Eastern and Western Rapti, and Babai Chepang River, reached dangerous levels.
5. 4 bridges were fully damaged in Koshi province, including the Bailey Bridge over Hewa Khola on Mechi Highway in Panchthar and a permanent bridge over Rati Khola connecting Ilam Sandakpur due to flooding.
6. According to the preliminary report by the Nepal Electricity Authority, a total of 22 hydroelectric power plants and 9 transmission lines have been damaged, resulting in the disruption of 1,000 MW of power generation, including 456 MW from the Upper Tamakoshi Hydropower Plant.
7. Across the country, 7,000 hectares of paddy fields have been reported inundated, with significant livestock losses. Further details are awaited.
8. Massive Landslide at Jhyaplekhola of Dhading District buried 4 vehicles including two tourist buses, a passenger bus and a van resulting in the death of 35 people.
9. According to a preliminary report by the Health Emergency Operation Center, 10 health facilities have been damaged in the incident.

Situation Overview

Continuous rainfall from 27th September, 2024 resulted in multiple incidents of flood and inundations, small, medium and large scale landslides across the country. According to the Department of Hydrology and Meteorology (DHM), out of the 222 measurement stations where rainfall was simultaneously recorded across the country, 77 stations reported heavy rainfall, with amounts exceeding 200 mm on Saturday.

Among the hydrological gauging stations installed in various rivers across the country, 23 stations recorded water levels surpassing the danger level, and an additional 14 stations recorded water levels exceeding the warning level. The water flow in the Saptakoshi River has reached the highest level in 56 years, The water discharge of Saptakoshi river is 6,43,040 cusec.

After 22 years, Kathmandu witnessed record-breaking rainfall on Friday and Saturday. DHM reported that the rainfall on Saturday was the highest ever in the valley's recorded history, surpassing the previous record set in 2002. Rainfall at the Tribhuvan International Airport station in Kathmandu reached 239.7 millimeters in 24 hours, compared to the previous record of 177 millimeters in 2002.



Displaced HH in Affected Districts

S.N.	Province	District	Displaced HHs
1	Bagmati Province	Dolakha	6
2		Ramechhap	2
3		Sindhuli	12
4		Kavrepalanchok	30
5		Sindhupalchok	7
6		Rasuwa	2
7		Nuwakot	1
8		Dhading	23
9		Makwanpur	8
10		Bhaktapur	4
11		Lalitpur	35
12		Kathmandu	34
13	Koshi Province	Dhankuta	17
14		Ilam	11
15		Panchthar	3
16		Morang	55
17		Jhapa	243
18		Udayapur	37
19		Bhojpur	5
20		Terhathum	1
21		Khotang	2
22		Okhaldhung	8
23		Solukhumbu	2
Total			548

Damage on Bridges

S.No	Type of Bridge	Unit	Quantity
1	Culvert	Nos	1
2	Wooden Bridge	Nos	0
3	Bailey Bridge	Nos	2
4	RCC Simply Supported	Nos	4
5	RCC Prestressed	Nos	2
6	Arch	Nos	1
7	Steel Truss Bridge	Nos	1
8	Trail bridge	Nos	2
9	Others	Nos	0
10	Total length of bridge affected	Meter	610
11	Estimated cost of preliminary damage (in thousand NPR)	Amount	408,000.00

Damaged Highways and Roads

S.N.	Description	Unit	Quantity
1	Total number of highways/roads affected by blockages	Number	42
2	Total number of blockage points	Number	784
3	Locations with both-way traffic restored	Number	333
4	Locations with one-way traffic operational	Number	396
5	Number of highways currently blocked	Number	19
6	Number of locations currently blocked	Number	56
7	Total time highways remained blocked (hours)	Hours	236,552.61
8	Total length of highways affected	meters	21500
9	Estimated cost of preliminary damage (in thousand NPR)	Amount	164,540.00

Damage on Irrigation

Details	Estimated Loss
10 branch canals under the Sunsari-Morang Irrigation Project	NPR 50 million
Bagmati Irrigation Project eastern and western main canals	NPR 54 million
Jhajh Irrigation canal Project	NPR 30 million
Koshi Pump-Chandra Canal Project	NPR 5 million
Kamala Irrigation Management Project	NPR 50 million
Narayani Irrigation Management Project	NPR 1 million
Mahakali Irrigation Project in Kanchanpur	NPR 1.14 billion
Total estimated loss and damage	NPR 190 million

The damage to the Sunkoshi-Marin Diversion Multipurpose Project yet to be estimated.

Key Actions Taken

On September 26, the NDRRMA issued a High Alert Notice to inform the public about the potential risks associated with heavy rainfall forecasted for September 27-28 by the DHM. As a precaution, night bus services on highways in affected districts were halted for safety reasons.

On September 28, the Command Post Meeting chaired by the Minister for Home Affairs assessed the situation and discussed strategies for effective rescue and relief. This meeting included the Home Secretary and security chiefs. Additionally, an Emergency Meeting led by the Acting Prime Minister provided further directives for a more coordinated response to assist those affected by floods and landslides.

On September 29, an all-party meeting chaired by the Acting Prime Minister briefed political parties on the recent loss and damage across the country and the government's ongoing rescue and relief efforts. The government called for coordinated efforts from all parties to ensure an effective response.

On September 29, a cluster co-lead meeting held at NDRRMA evaluated immediate needs and called for collaboration with humanitarian organizations to enhance the effectiveness of the response and relief efforts.

Immediate Needs and Priorities

- Temporary shelter for displaced families.
- Tents with light and other utilities.
- Emergency Water and Sanitation.
- Rescue and relief for those stranded on highways due to flash floods and landslides.
- Provision of food and other humanitarian supplies.
- Restoration of communication networks and electricity in affected areas.
- Construction of temporary learning centers and health facilities.
- Debris clearance and repair of disrupted road networks and highways.

Other Needs

- Prefabricated steel truss vehicular bridge – Total of 1050 meters
- Outboard motor with raft – 50 sets
- Heavy lifting drones (for relief material distribution) – 25 Unit
- Combi tools for road traffic accident (hydraulic) – 60 sets
- Flood light tower (Portable) – 100 units
- Power ascender with rope launcher – 30 sets



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